Chapter 02

- $A\cos(2\pi(440)t + \varphi)$ — radian frequency (ω)

  A: amplitude, 440: frequency (H_z), φ: phase

  $2\pi f t = \omega t$

period $= \frac{2\pi}{\omega} = \frac{1}{f}$, peak when $\omega t - \varphi = 0$ (코사인은 0일때 peak)

- time-shift

phase 변화에 t를 $t - t_m$으로 대치하여 표현

$A\cos(\omega(t - t_m)) = A\cos(\omega t + \varphi)$

$\therefore -\omega t_m = \varphi$, $t_m = -\frac{\varphi}{\omega}$

- Complex Exponentials

$z(t) = Xe^{j\omega t}$ — complex Exponentials

$X = Ae^{j\varphi}$

complex amplitude

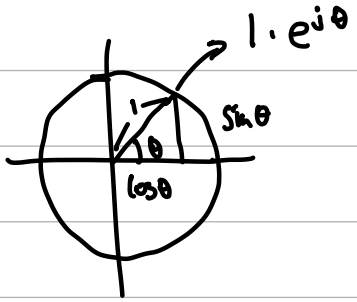
- Polar Coordinate

$A e^{j\theta}$

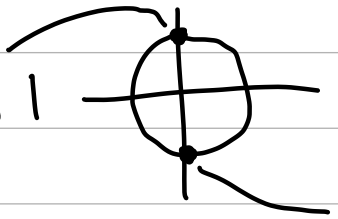
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크기 $\cos\theta + j\sin\theta$

real imaginary



$e^{j\frac{\pi}{2}} = j = 0 + j1$

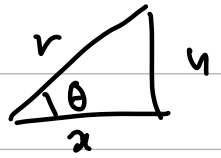


$e^{j\pi} = -1$

$re^{j\theta} = x + jy = r(\cos\theta + j\sin\theta)$

$x = r\cos\theta$

$y = r\sin\theta$



역탄젠트 계산 $\theta = \tan^{-1}\left(\frac{y}{x}\right)$

- $e^{jwt} = \cos(wt) + j\sin(wt)$

$\therefore \cos(wt) =$ real part of e^{jwt}

$\therefore A\cos(wt+\varphi) =$ real part of $\begin{cases} Ae^{j(wt+\varphi)} \\ \| \\ Ae^{j\varphi} \cdot e^{jwt} \end{cases}$

- Complex Amplitude (phasor)

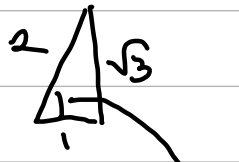
$X = Ae^{j\varphi}$ $\quad \therefore A\cos(\omega t + \varphi) = \text{real part of } Xe^{j\omega t}$

- Phasor Addition

$1e^{j0} \rightarrow 1\cos 0 + j \cdot 1 \sin 0 = 1 + j0$

$\sqrt{3}\, e^{j\,0.5\pi} \rightarrow \sqrt{3}\cos\left(\frac{\pi}{2}\right) + j\sqrt{3}\sin\left(\frac{\pi}{2}\right) = 0 + j\sqrt{3}$

$1 + j0 + 0 + j\sqrt{3} = 1 + j\sqrt{3}$



$\arctan\left(\frac{\sqrt{3}}{1}\right) = 60° = \frac{\pi}{3}$

$\therefore 1e^{j0} + \sqrt{3}e^{j0.5\pi} = 2e^{j\frac{\pi}{3}}$

$\tan\theta$	
0	0
$\frac{\sqrt{3}}{3}$	30
1	46
$\sqrt{3}$	60